

Emergence Dataset #1:

Anisotropy, Isomer Stability, and Fall Transitions

VSEPR-SIM Development

Version 1.0.0

Abstract

Emergence Dataset #1: Anisotropy and Isomer-Fall Transitions is a structured benchmark for directional-response and state-transition phenomena across molecular, cluster, and coarse-grained systems. The dataset records initial and final structural states, anisotropy descriptors, driving perturbations, transition barriers, persistence times, and fall-event classifications. Its purpose is to support deterministic simulation studies of symmetry breaking, metastability, conformational switching, field-induced transitions, and anisotropy-driven collapse events within the VSEPR-SIM architecture.

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1 Core Scientific Idea

The dataset captures two coupled behaviours:

1.1 Anisotropy

A system has direction-dependent behaviour. This can be mechanical, electrical, optical, geometric, transport-related, or energetic. Anisotropy is a standard property class in materials science and molecular systems.

1.2 Isomer Fall

Isomer Fall := a discrete transition from one local structural state to another under perturbation.

That perturbation may be thermal, field-driven, flow-driven, collision-driven, stress-driven, radiation-driven, or concentration-driven.

The dataset backbone is:

$$\boxed{\text{State} \rightarrow \text{Perturbation} \rightarrow \text{Directional Response} \rightarrow \text{Structural Transition}} \quad (1)$$

2 Dataset Organisation

Subset	Name	Scope
ED1A	Molecular	Small molecules, torsions, ring flips, cis/trans
ED1B	Coordination	Metal centres, ligand distortions, actinide assemblies
ED1C	Bead/Structure	Directional collapse, branch reorientation, field-driven deformation

This prevents butane and plutonium-ligand bead assemblies from occupying the same row.

3 Isomer-Fall Detection Criteria

A sample is tagged as an isomer-fall event if *any* of the following conditions hold:

$$\Delta E_{\text{barrier}} < E_{\text{perturb}} \quad (2)$$

$$\text{RMSD}(X_t, X_{\text{ref}}) > \tau_{\text{geom}} \quad (3)$$

$$q_{\text{state}}(t + \Delta t) \neq q_{\text{state}}(t) \quad (4)$$

where:

- $\Delta E_{\text{barrier}}$ = barrier to alternate configuration
- E_{perturb} = supplied perturbation energy
- RMSD = structural displacement (root-mean-square deviation)

- τ_{geom} = geometric displacement threshold ($\text{\AA}$)
- q_{state} = discrete isomer/state label

A fall is not just motion. It is a **state transition**.

3.1 Implementation

Listing 1: Fall detection in C++ (emergence_dataset.hpp)

```
struct FallDetectionConfig {
    double rmsd_threshold{0.5};      // tau_geom (A)
    double barrier_ratio_min{1.0};    // threshold for barrier criterion
    bool   check_state_label{true};   // check q_state change?
};
```

4 Anisotropy Descriptors

Anisotropy is not a vibe. It is a quantified ratio.

4.1 Geometric Anisotropy

From principal eigenvalues of the gyration tensor:

$$A_{\text{geom}} = \frac{\lambda_1}{\lambda_3}, \quad \lambda_1 \geq \lambda_2 \geq \lambda_3 \quad (5)$$

Asphericity:

$$\kappa = 1 - \frac{3(\lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1)}{(\lambda_1 + \lambda_2 + \lambda_3)^2} \quad (6)$$

4.2 Mechanical Anisotropy

$$A_{\text{mech}} = \frac{P_{\text{max}}}{P_{\text{min}}} \quad (7)$$

for some property P (modulus, yield response, or compliance).

4.3 Transport Anisotropy

$$A_{\text{trans}} = \frac{D_{\parallel}}{D_{\perp}} \quad \text{or} \quad A_{\sigma} = \frac{\sigma_{\parallel}}{\sigma_{\perp}} \quad (8)$$

4.4 Field Anisotropy

$$A_E = \frac{E_{\text{max},\parallel}}{E_{\text{max},\perp}} \quad (9)$$

4.5 Optical/Tensor Anisotropy

$$A_n = \frac{n_{\parallel}}{n_{\perp}} \quad (10)$$

All ratios are ≥ 1.0 for anisotropic systems. A ratio of 1.0 means perfectly isotropic.

5 Fall Modes

Categorical labels for the transition type:

Mode	Description
rotation_fall	Rigid-body reorientation
torsion_fall	Dihedral angle transition (e.g. anti $\rightarrow$ gauche)
coordination_fall	Change in coordination number or shell geometry
buckling_fall	Structural buckling under load
collapse_fall	Compaction / directional collapse
ligand_swap_fall	Exchange of ligand positions
electrostatic_reorientation	Field-driven dipole realignment
thermal_isomerization	Heat-driven isomer switching
field_induced_isomerization	External field forces state change
surface_binding_fall	Adsorption-driven reconfiguration
conformer_fall	Ring flip, chair/boat, etc.
structural_fall	General structural rearrangement
orientation_fall	Facet or axis reorientation

6 Fall Severity Score

$$S_f = w_1 \frac{\text{RMSD}}{\text{RMSD}_{\max}} + w_2 \frac{|\Delta E|}{|\Delta E|_{\max}} + w_3 \frac{\tau_{\text{persist}}}{\tau_{\max}} \quad (11)$$

where $S_f \in [0, 1]$ and the default weights are:

w_1	w_2	w_3
0.4	0.4	0.2

Normalisation constants: $\text{RMSD}_{\max} = 5.0 \text{ \AA}$, $|\Delta E|_{\max} = 50.0 \text{ kcal/mol}$, $\tau_{\max} = 1000 \text{ ps}$.

7 Dataset Schema

7.1 Minimal Schema

$$X_i = (\text{id}, \text{class}, C, s_0, s_1, \vec{A}, d, |d|, \tau, \Delta E, \Delta G, \text{RMSD}, f, m) \quad (12)$$

7.2 Full CSV Header

```
sample_id,family,system_class,composition,charge_state,mass,
initial_state,final_state,transition_class,
anisotropy_type,anisotropy_ratio,
axis_1_x,axis_1_y,axis_1_z,axis_2_x,axis_2_y,axis_2_z,
driver_type,driver_magnitude,temperature_K,pressure_atm,medium,
barrier_energy_kcal,deltaE_kcal,deltaG_kcal,rmsd_A,
residence_time_init_ps,residence_time_final_ps,
fall_flag,fall_mode,fall_severity,
recoverable,hysteresis,metastable,notes
```

7.3 Recommended File Split

For larger datasets, split into:

- `state.csv` — identity, states, labels
- `event.csv` — fall flags, modes, severity
- `tensor.csv` — anisotropy eigenvalues, axes, ratios
- `environment.csv` — drivers, T, P, medium

8 Label System

8.1 Binary Labels

Label	Meaning
<code>fall_flag</code>	0/1 — did an isomer fall occur?
<code>recoverable_flag</code>	0/1 — can the system return to initial state?
<code>metastable_flag</code>	0/1 — was the initial state metastable?

8.2 Multiclass Labels

`anisotropy_type`, `fall_mode`, `driver_type`

8.3 Continuous Labels

Anisotropy ratios, barrier energy, RMSD, persistence time, directional tensor eigenvalues.

This provides both supervised ML labels and physically interpretable descriptors.

9 Benchmark Exemplars

9.1 Example 1: Butane (anti $\rightarrow$ gauche)

Field	Value
Family	ED1A-Molecular
Composition	C ₄ H ₁₀
Initial state	anti
Final state	gauche
Trigger	thermal
Barrier	3.4 kcal/mol
ΔE	0.9 kcal/mol
RMSD	0.8 Å
Fall mode	<code>torsion_fall</code>
Recoverable	yes

9.2 Example 2: Cyclohexane (chair $\rightarrow$ boat)

Field	Value
Family	ED1A-Molecular
Composition	C_6H_{12}
Initial state	chair
Final state	boat
Barrier	10.8 kcal/mol
ΔE	5.5 kcal/mol
Fall mode	<code>conformer_fall</code>
Metastable	yes (boat is metastable)

9.3 Example 3: Organometallic Coordination Shell (trans $\rightarrow$ cis)

Field	Value
Family	ED1B-Coordination
Composition	$PtCl_2(NH_3)_2$
Initial state	trans (transplatin)
Final state	cis (cisplatin)
Trigger	electric field
Fall mode	<code>coordination_fall</code>
Hysteresis	yes

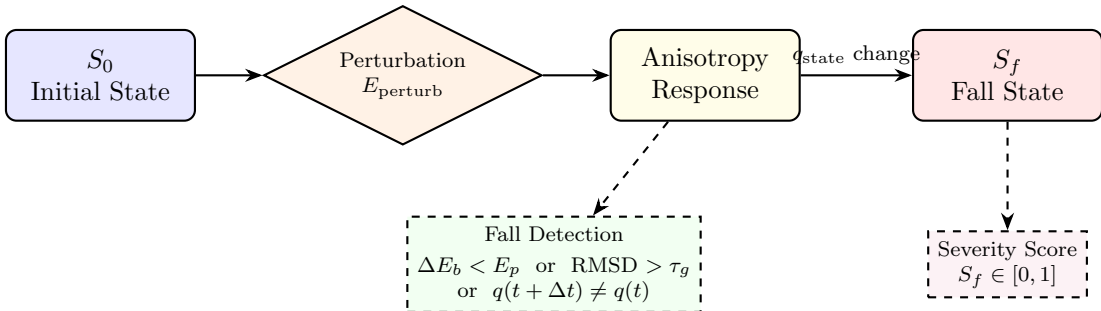
9.4 Example 4: Anisotropic Crystal Platelet

Field	Value
Family	ED1C-Bead/Structure
Composition	MoS_2 fragment
A_{mech}	8.5
Trigger	shear stress
Fall mode	<code>orientation_fall</code>
Recoverable	yes
Hysteresis	yes

9.5 Example 5: Bead-Assembled Structure

Field	Value
Family	ED1C-Bead/Structure
Composition	CG_branch_12bead
Initial state	branch_aligned
Final state	collapsed
A_{geom}	4.2
ΔE	-12.0 kcal/mol
Fall mode	<code>collapse_fall</code>
Recoverable	no (irreversible compaction)

10 Transition Diagram



11 Integration with VSEPR-SIM

11.1 Mapping to Existing Infrastructure

ED1 Concept	VSEPR-SIM Component	Header
State	atomistic::State / StateC	state.hpp / state.c.hpp
Anisotropy tensor	InertiaFrame	inertia_frame.hpp
RMSD	compute_rmsd()	alignment.hpp
Conformer search	ConformerFinder	conformer_finder.hpp
Fall event	EventRecord in Ω	state.c.hpp
Provenance	SourceTag (Σ)	state.c.hpp
Environment	Environment ($\mathcal{E}$)	state.c.hpp
Perturbation	PerturbationDriver	emergence_dataset.hpp

11.2 Relationship to Dev Day ~42 Notation

Each emergence sample maps directly to the revised notation:

- S_0 : initial state of the sample (StateC with composition, environment, constraints)
- $T(S_0, t, \Lambda)$: perturbation-driven transformation
- S_f : outcome state after fall (or non-fall)
- Ω : event log recording the fall event
- Q : quality flags indicating whether the fall is trustworthy

11.3 Relationship to Deep Research Engine

The deep research engine (Phase 10A – 41A) can feed samples into ED1:

- Phase 10A enumerated geometries become ED1A molecular samples
- Phase 20B bead formations become ED1C bead/structure samples
- SHA-512 pathway hashes provide provenance (Σ) for each sample

12 Project Tree

```

atomistic/
  core/
    state.hpp           <-- Particle-level state
    state_c.hpp         <-- Dev Day ~42 revised notation
    alignment.hpp       <-- RMSD, Kabsch alignment
  datasets/
    emergence_dataset.hpp <-- THIS FILE (ED1 schema + detection + export)
  research/
    deep_research.hpp   <-- 4-phase pipeline (feeds ED1)

coarse_grain/
  core/
    inertia_frame.hpp   <-- Principal-axis decomposition (anisotropy source)
    bead.hpp            <-- Bead struct (C = Bead representation)

src/sim/
  conformer_finder.hpp  <-- Isomer/conformer enumeration

tests/
  test_emergence_dataset.cpp <-- 25 tests (enums, detection, severity, exemplars)

docs/
  section_emergence_dataset_1.tex <-- THIS DOCUMENT

```

13 Summary

Emergence Dataset #1 establishes:

1. **Three subfamilies** (ED1A Molecular, ED1B Coordination, ED1C Bead/Structure) preventing forced coexistence of incompatible system types.
2. **Strict fall detection** via three independent criteria: barrier energy, RMSD displacement, and state-label change.
3. **Quantified anisotropy** through eigenvalue ratios of relevant tensors (geometric, mechanical, transport, field, optical).
4. **Categorical fall modes** with a composite severity score $S_f \in [0, 1]$.
5. **Five benchmark exemplars** spanning butane torsion, cyclohexane ring flip, transplatin isomerisation, crystal platelet reorientation, and bead-assembly collapse.
6. **CSV export** with a full schema header for both single-file and split-file workflows.
7. **Direct integration** with the VSEPR-SIM state machine ($S_0 \rightarrow T \rightarrow S_f$), provenance system (Σ), event log (Ω), and quality flags (Q).

The C++ implementation lives in `atomistic/datasets/emergence_dataset.hpp` and is validated by 25 tests covering all schema components, detection logic, anisotropy computation, severity scoring, and benchmark exemplars.